

# Jake Joyner

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Engineer and founder, building tamper-evident audit infrastructure for AI inference. Strongest in Rust, edge compute, cryptographic provenance, and EU AI Act compliance. Currently solo on AILedger; in early customer conversations.

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## SELECTED WORK

### AILedger – Founder

2025-present · [ailedger.dev](https://ailedger.dev)

Tamper-evident chain of custody for every AI inference an enterprise's product makes, designed for the EU AI Act Article 12 conformance question (Article 79 enforcement August 2, 2026).

- Cloudflare Workers proxy with input/output SHA-256 hashing and RFC 8785 JCS canonicalization
- Per-tenant audit chain with deterministic verification path
- Customer discovery underway with EU-exposed enterprise AI teams

### skillrt – Rust, published

[crates.io/crates/skillrt](https://crates.io/crates/skillrt)

Skill-runtime primitive for AI agents: portable skill-pack format, dynamic loading, dispatch.

### matrixrt – Internal runtime

2026

Dynamic task→skill matching runtime. Coverage-based scoring with  $\epsilon$ -greedy exploration; outcomes update per-skill priors over an append-only ledger. Designed for high-throughput operation.

### Multi-silo collaboration infrastructure – Internal

2026

Cross-agent work isolation, audit logging, and signaling layer (Plaza / Border / Overseer pattern). Built on Python stdlib, tailscale, and systemd user units.

## EARLIER EXPERIENCE

### Transfer Research Experience (T-REX), UC San Diego

August 2025

Paired with faculty mentor; studied optimization techniques in modern machine learning. Presented research poster.

### Restoration Technician, True West Motorworks

2025

Diagnostics, engine repair, and spot welding on neglected vehicles, primarily air-cooled VWs.

## EDUCATION

### University of California, San Diego

September 2025 – April 2026 (departed)

Mathematics / Computer Science · 4.0 GPA · Departed to focus full-time on AILedger.

### Santa Monica College

February 2023 – June 2025

Computer Science · 3.9 GPA · Scholars Program · Dean's List  
Coursework: abstract algebra, combinatorics, extremal graph theory.

## TECHNICAL SKILLS

**Languages:** Rust · TypeScript · Python · Go · JavaScript · C/C++ · SQL · Java

**Infrastructure:** Cloudflare Workers · edge compute · audit-trail systems · cryptographic provenance (SHA-256, RFC 8785 JCS) · AI inference middleware · PostgreSQL · SQLite · Docker

**Frontend:** React · HTML / CSS

**Tools:** Linux · Vim · Git · LaTeX · OpenGL

**Domain knowledge:** EU AI Act (Articles 12 & 79) · tamper-evident logging · AI inference observability · compliance infrastructure

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